1 Samples correlation for different C

All figures for $N = 100$, $\langle z \rangle = 4$, $T = 1.0$, $p = 0.50$.

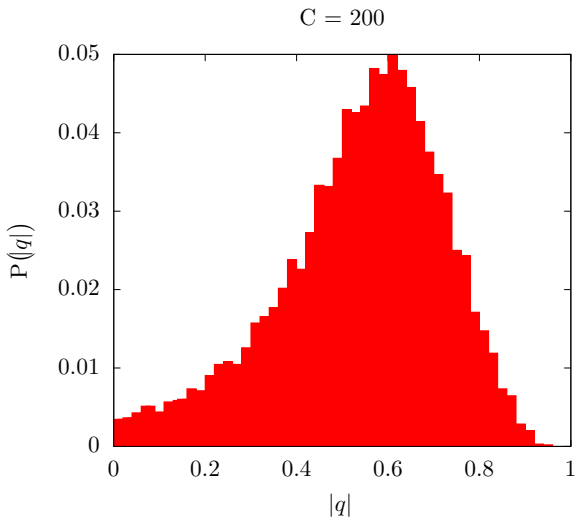


Figure 1.1

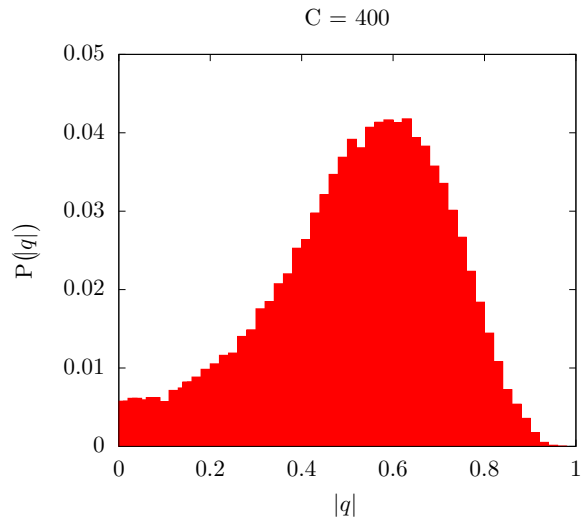


Figure 1.2

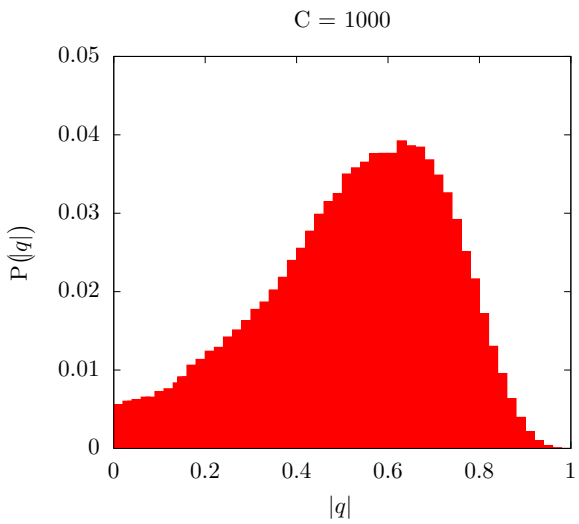


Figure 1.3

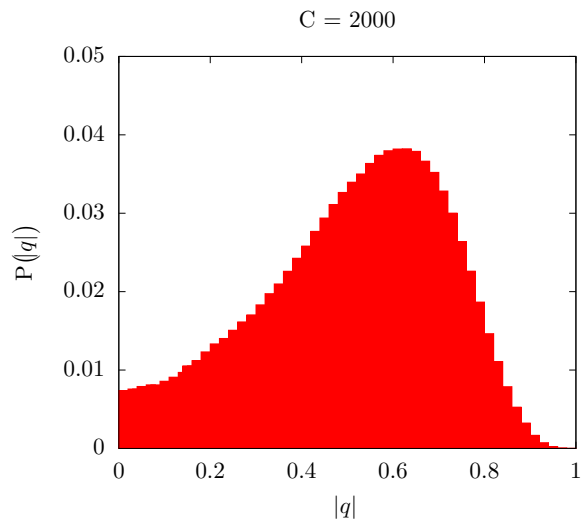


Figure 1.4

2 PL performance for different C

All figures for $N = 100$, $\langle z \rangle = 4$, $T = 1.0$, $p = 0.50$, $N_{\mathrm{g}} = 100$, $\tau = 2000$, $T_{\mathrm{F}}^{\mathrm{ini}} = 2.0$

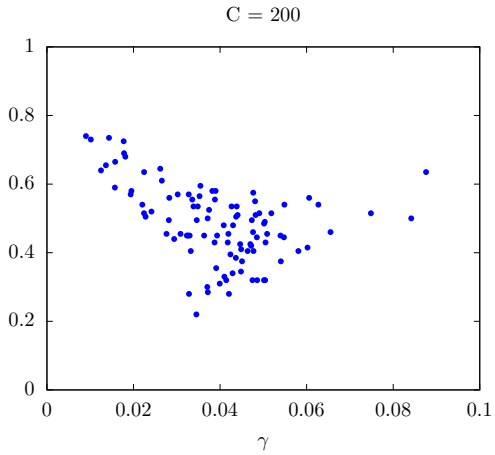


Figure 2.1

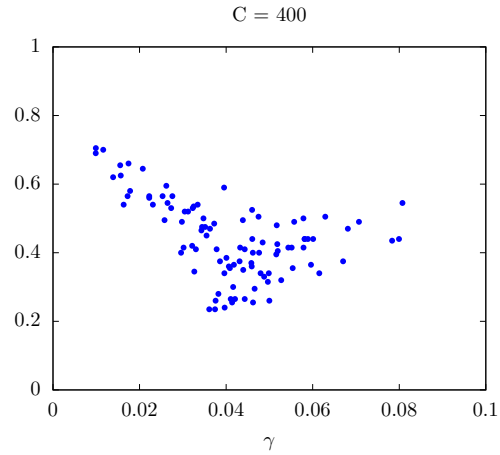


Figure 2.2

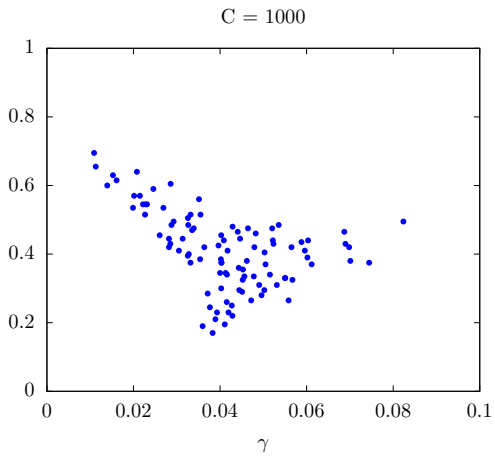


Figure 2.3

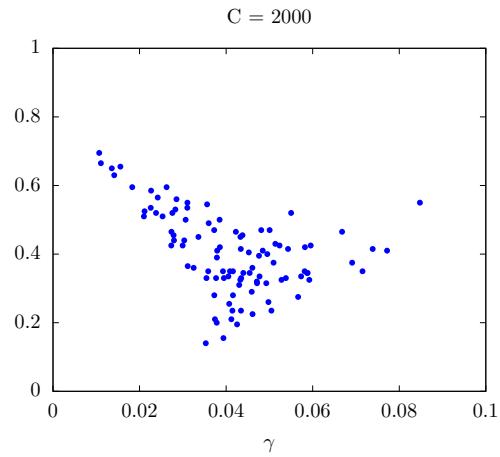


Figure 2.4

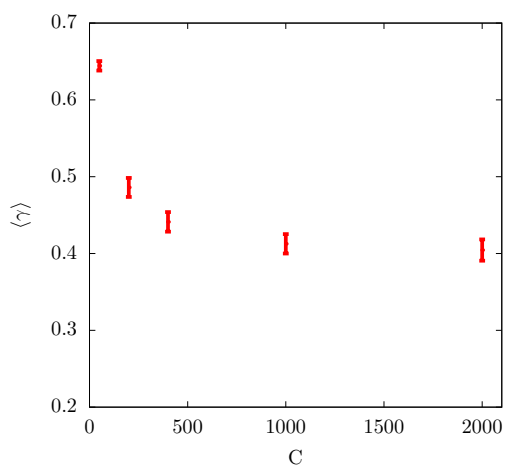


Figure 2.5: Dependency between average γ and the number of samples C.

3 PL performance for different τ

All figures for $N = 100$, $\langle z \rangle = 4$, $T = 1.0$, $p = 0.50$, $N_{\mathrm{g}} = 100$, $\mathrm{C} = 400$, $T_{\mathrm{F}}^{\mathrm{ini}} = 0.04$.

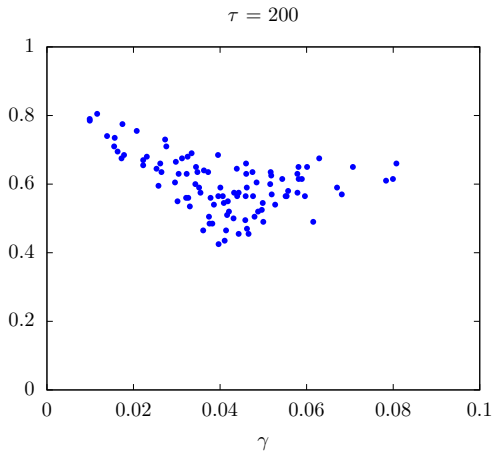


Figure 3.1

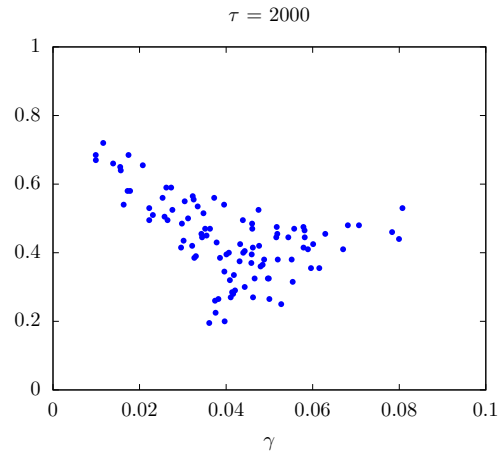


Figure 3.2

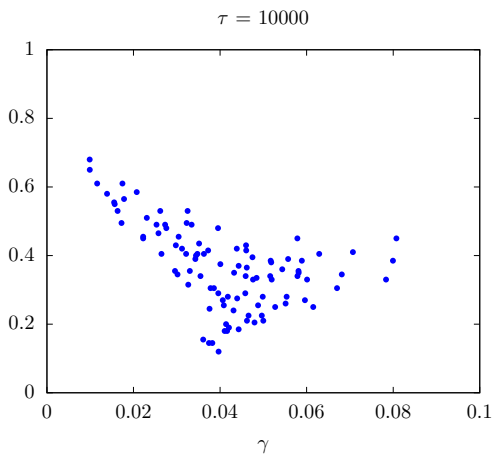


Figure 3.3

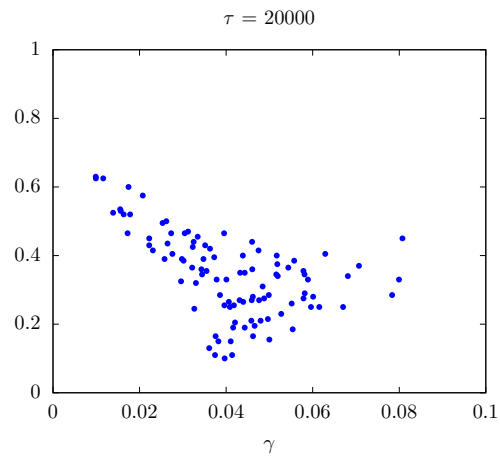


Figure 3.4

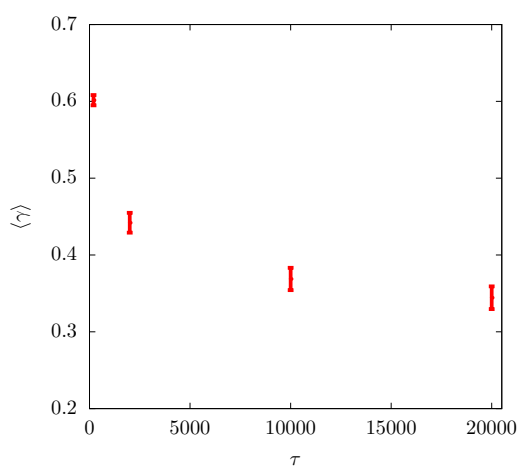


Figure 3.5: Dependency between average γ and the total number of Monte-Carlo steps τ.

4 Initial Fictitious Temperature effect on the PL performance

All figures for $N = 100$, $\langle z \rangle = 4$, $T = 1.0$, $p = 0.50$, $N_{\mathrm{g}} = 100$, $\mathrm{C} = 400$, $\tau = 1000$.

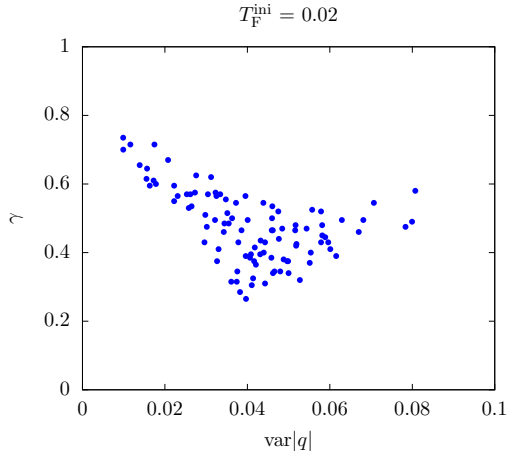


Figure 4.1

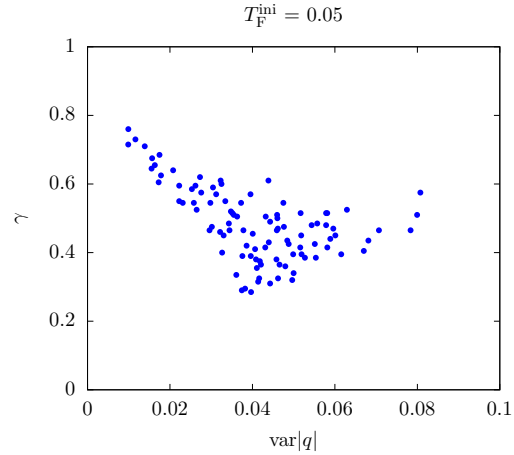


Figure 4.2

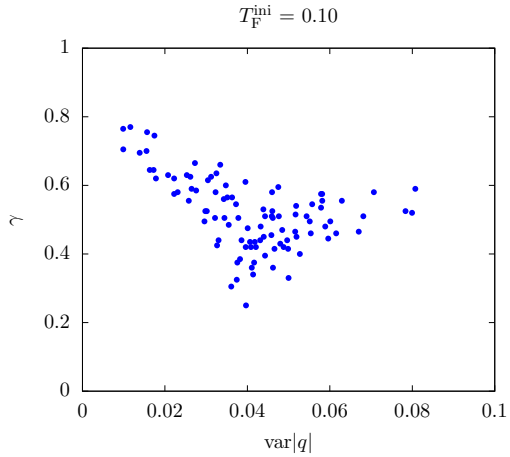


Figure 4.3

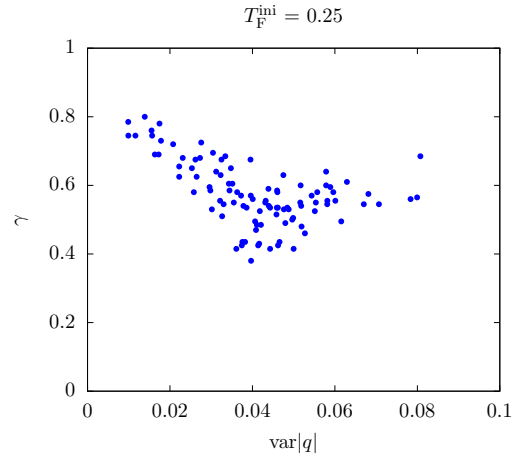


Figure 4.4

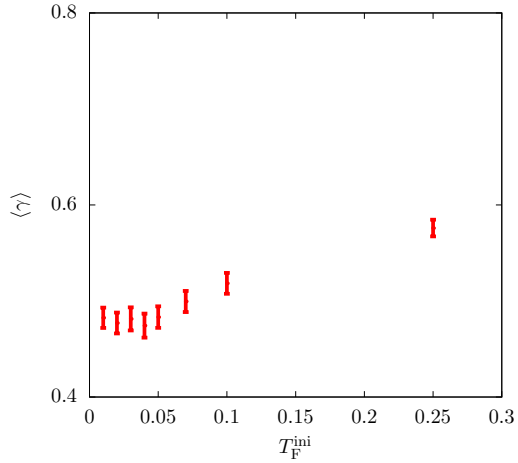


Figure 4.5: Dependency between average γ and the initial fictitious temperature of the simulated annealing.

5 Sample correlation distribution for all graphs with same p.

All figures for $N = 100$, $\langle z \rangle = 4$, $T = 1.0$, $N_{\mathrm{g}} = 100$, $\mathrm{C} = 1000$. Notice the vertical range difference for higher p values.

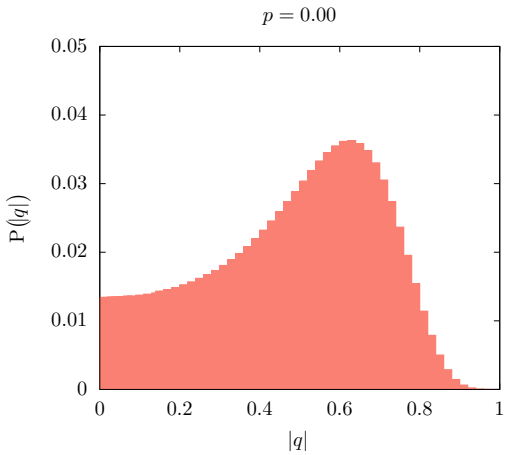


Figure 5.1

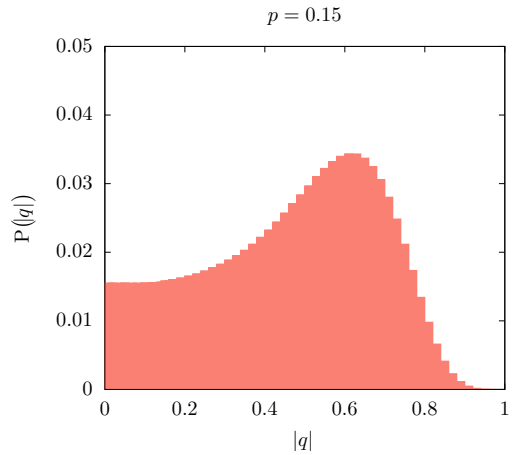


Figure 5.2

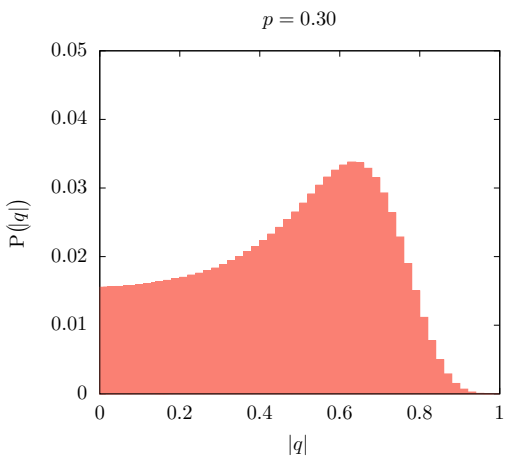


Figure 5.3

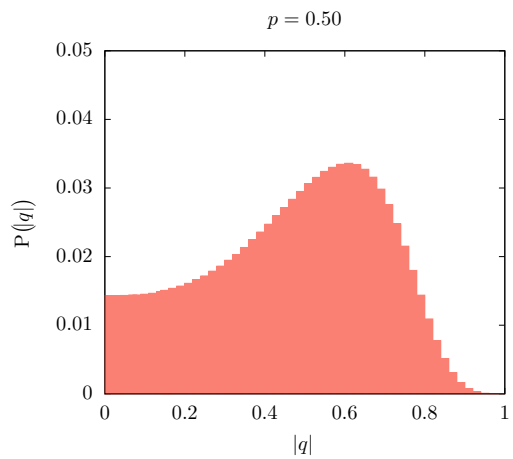


Figure 5.4

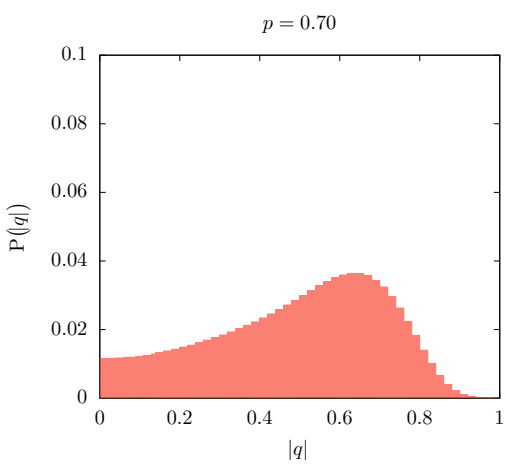


Figure 5.5

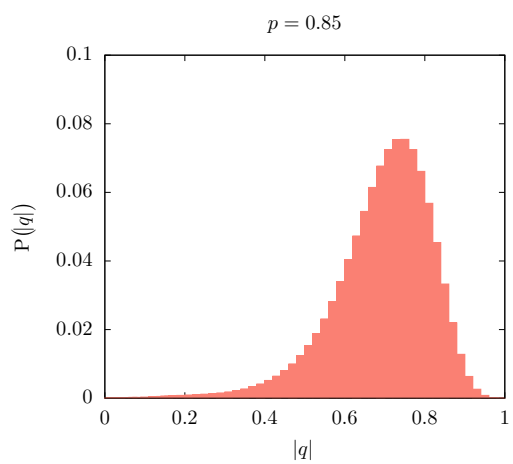


Figure 5.6

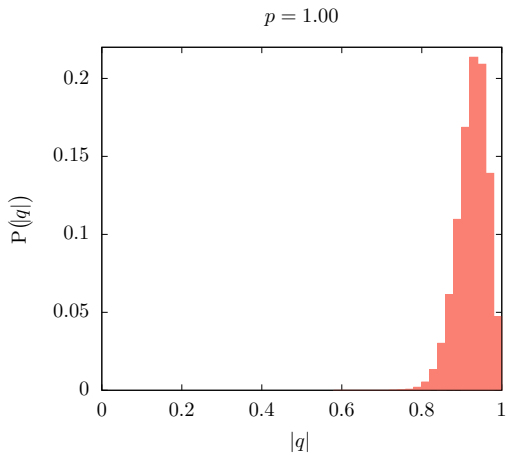


Figure 5.7

6 Sample correlation distribution for extreme PL performance graphs.

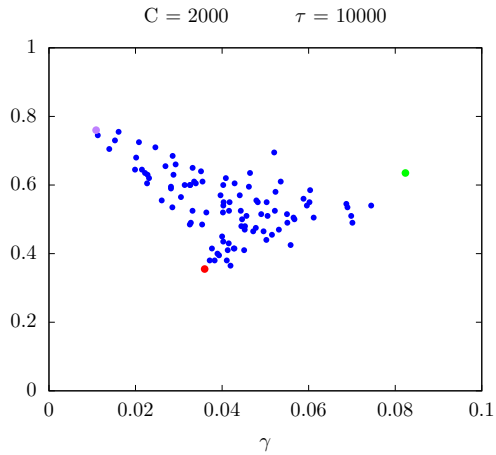


Figure 6.1: $N = 100$, $\langle z \rangle = 4$, $T = 1.0$, $p = 0.50$, $N_{\mathrm{g}} = 100$, $T_{\mathrm{F}}^{\mathrm{ini}} = 2.0$.

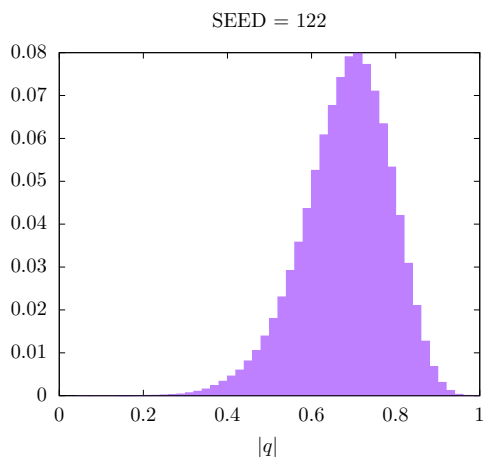


Figure 6.2: SEED with less var $|q|$ in Figure 6.1

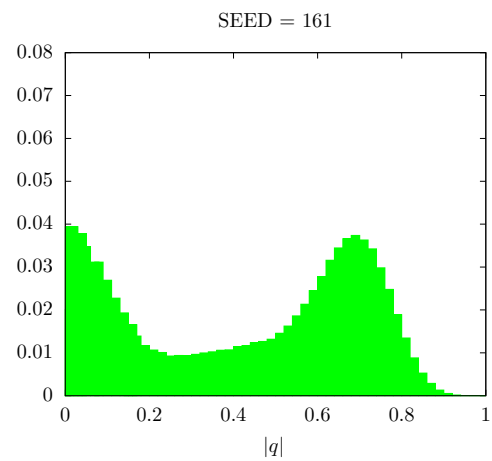


Figure 6.3: SEED with most var $|q|$ in Figure 6.1

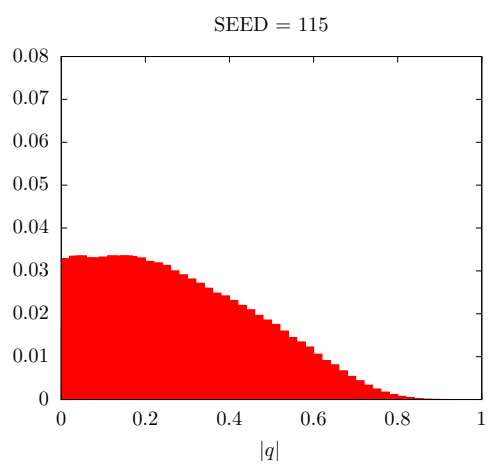


Figure 6.4: SEED with less γ in Figure 6.1

7 PL performance for C = 400, τ = 2000

All figures for $N = 100$, $\langle z \rangle = 4$, $T = 1.0$, $N_{\mathrm{g}} = 100$, $T_{\mathrm{F}}^{\mathrm{ini}} = 0.04$.

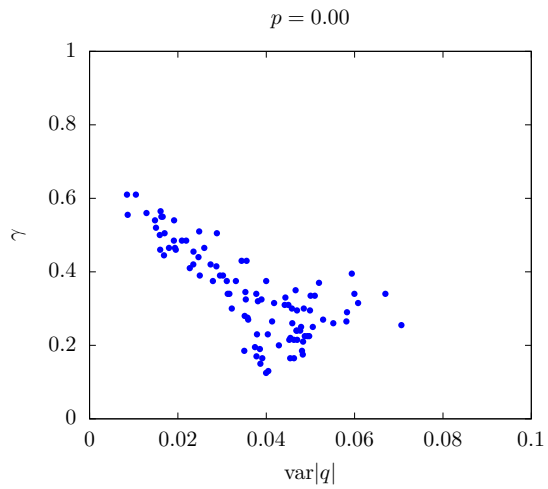


Figure 7.1

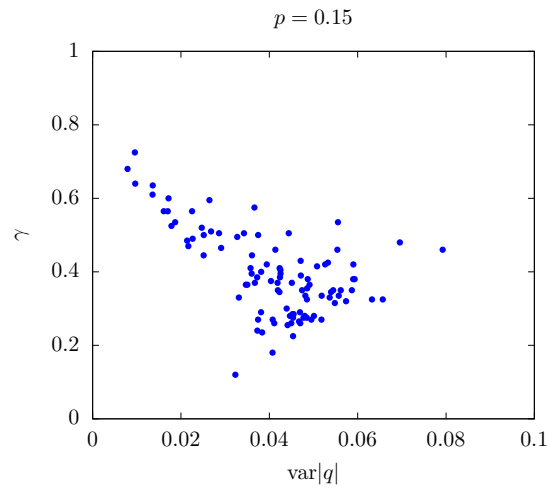


Figure 7.2

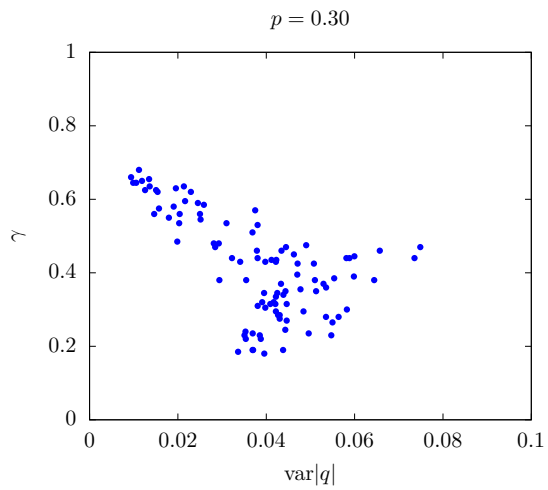


Figure 7.3

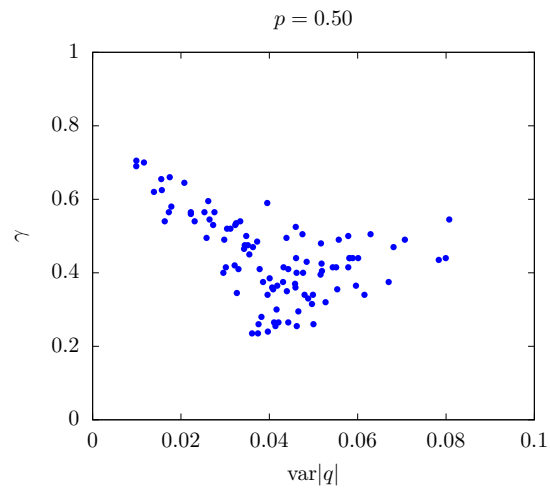


Figure 7.4

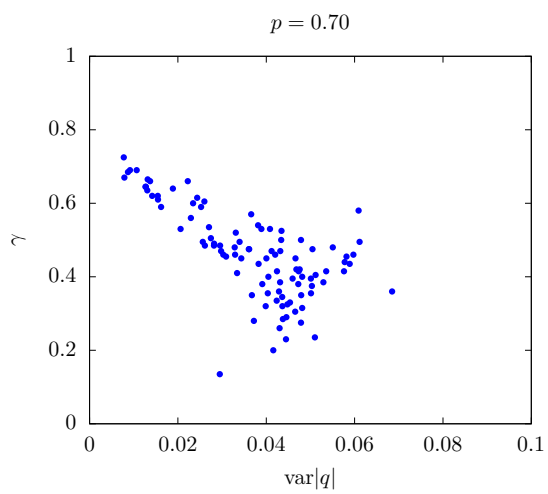


Figure 7.5

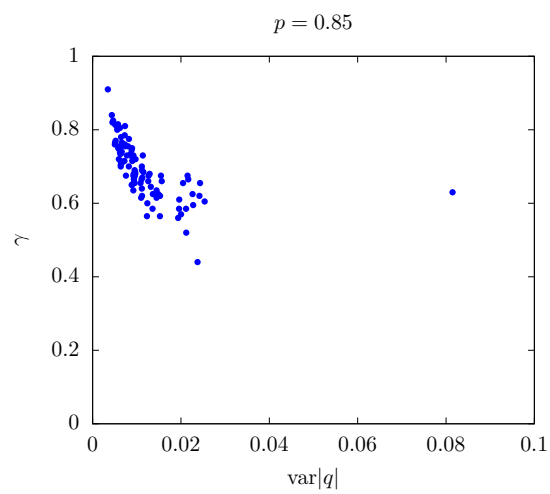


Figure 7.6

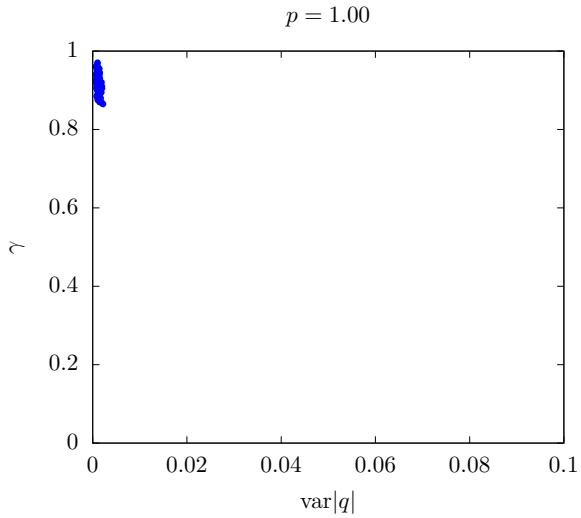


Figure 7.7

8 PL performance for different fictitious temperature reduction rhythms.

All figures for $N = 100$, $\langle z \rangle = 4$, $T = 1.0$, $N_{\mathrm{g}} = 100$, $T_{\mathrm{F}}^{\mathrm{ini}} = 0.04$, $\mathrm{C} = 400$, $\tau = 2000$.
($t \equiv$ current Monte-Carlo step)

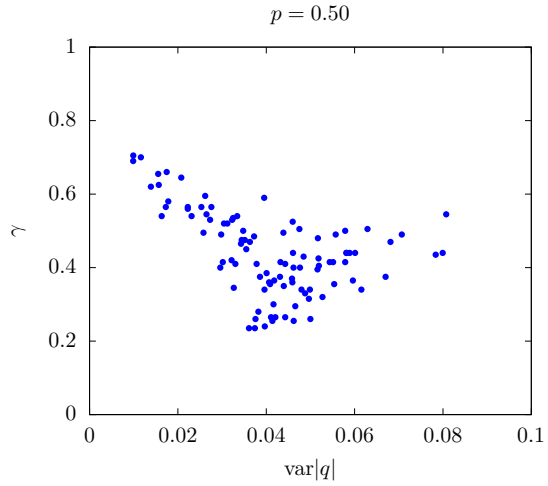


Figure 8.1: $T_{\mathrm{F}}(t) = T_{\mathrm{F}}^{\mathrm{ini}} \left(1 - \frac{t}{\tau}\right)$

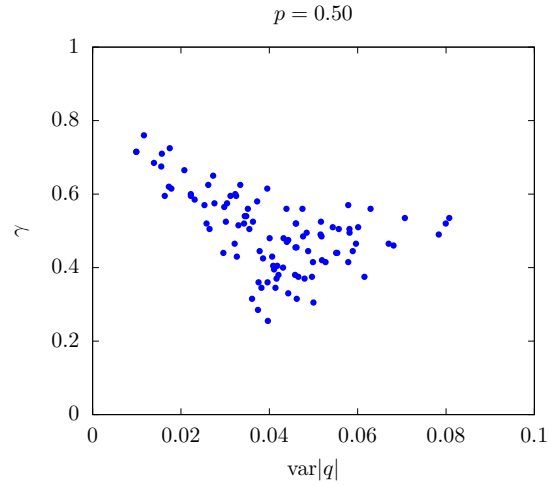


Figure 8.2: $T_{\mathrm{F}}(t) = \frac{T_{\mathrm{F}}^{\mathrm{ini}}}{\ln t} \left(1 - \frac{t}{\tau}\right)$

All the figures in this document are done with the Figure 8.1 equation.

9 Consistency of the PL performance

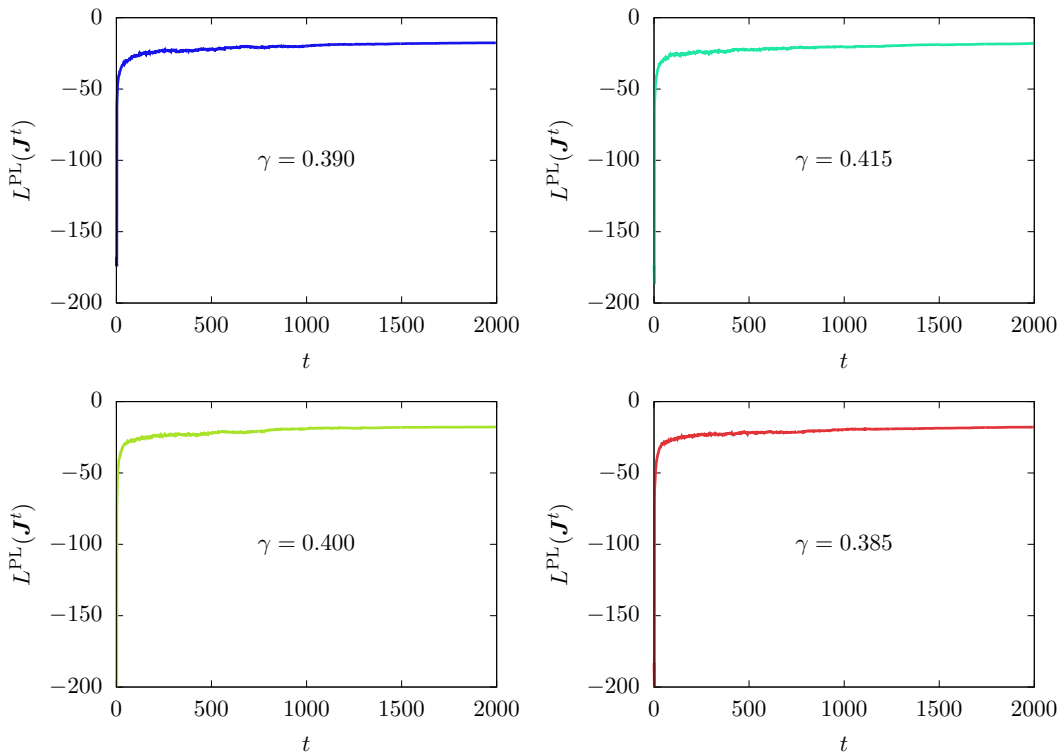


Figure 9.1: PL evolution in 4 different Monte-Carlo simulations for the SEED = 100 with $N = 100$, $\langle z \rangle = 4$, $p = 0.50$, $p = 1.0$, $T_{\mathrm{F}}^{\mathrm{ini}} = 0.04$, C = 400, $\tau = 2000$.

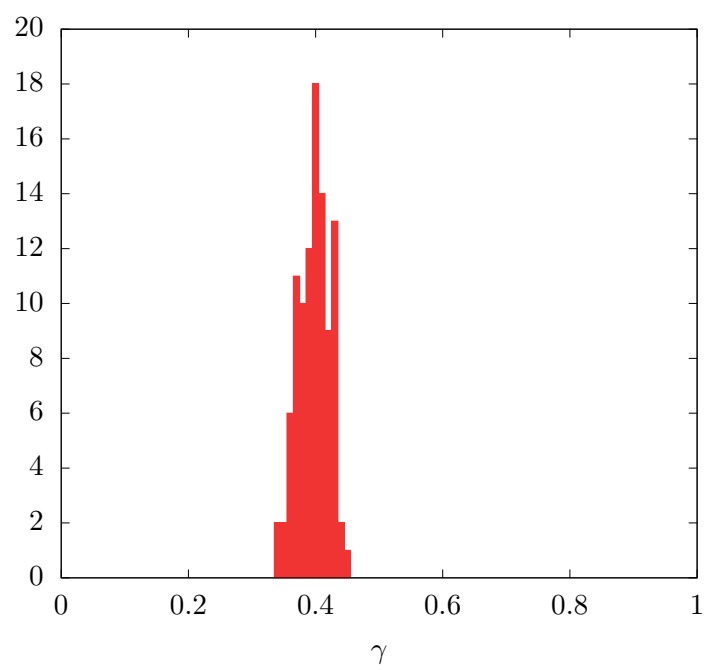


Figure 9.2: Gamma distribution in 100 different Monte-Carlo simulations for the SEED = 100 with $N = 100$, $\langle z \rangle = 4$, $p = 0.50$, $T = 1.0$, $T_{\mathrm{F}}^{\mathrm{ini}} = 0.04$, C = 400, $\tau = 2000$.

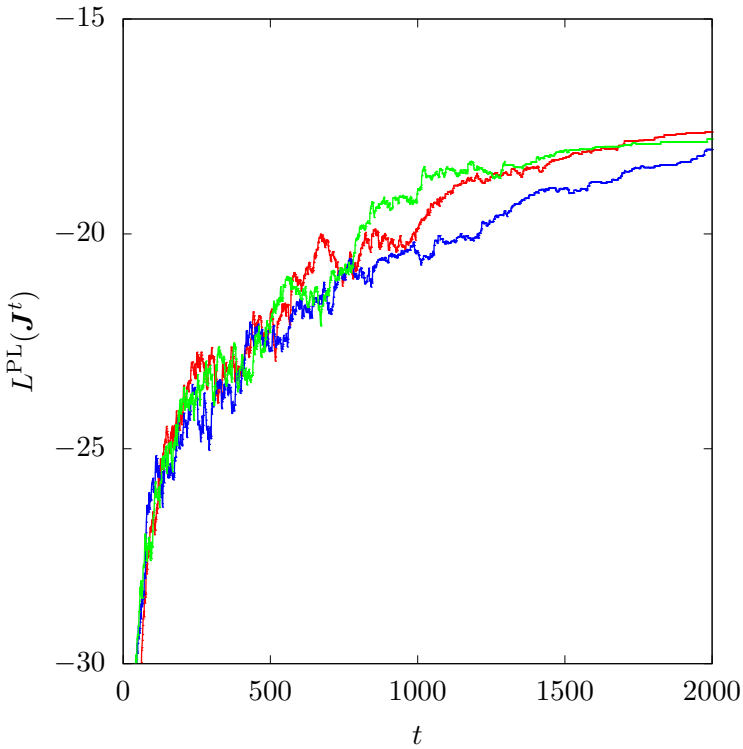


Figure 9.3: PL evolution in 3 different Monte-Carlo simulations zoomed in for the SEED = 100 with $N = 100$, $\langle z \rangle = 4$, $p = 0.50$, $T = 1.0$, $T_{\mathrm{F}}^{\mathrm{ini}} = 0.04$, C = 400, $\tau = 2000$.

10 Samples correlation for different T

All figures for $N = 100$, $C = 400$, $\langle z \rangle = 4$, $N_g = 100$.

$p = 0.00$ $p = 0.15$ $p = 0.30$ $p = 0.50$ $p = 0.70$ $p = 0.85$ $p = 1.00$

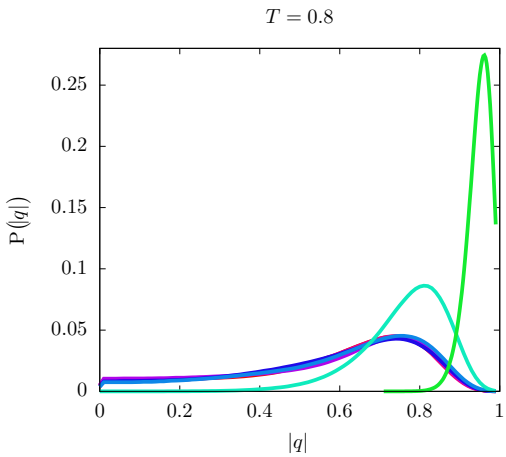


Figure 10.1

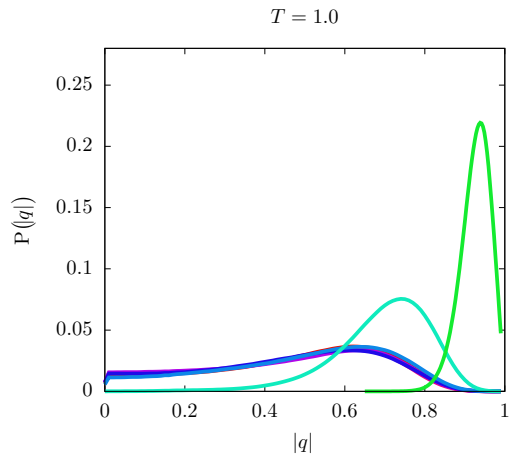


Figure 10.2

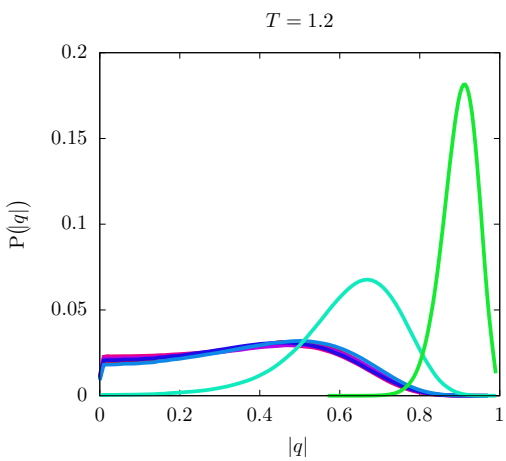


Figure 10.3

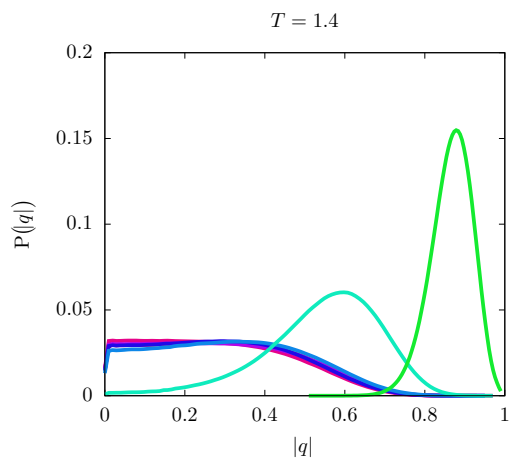


Figure 10.4

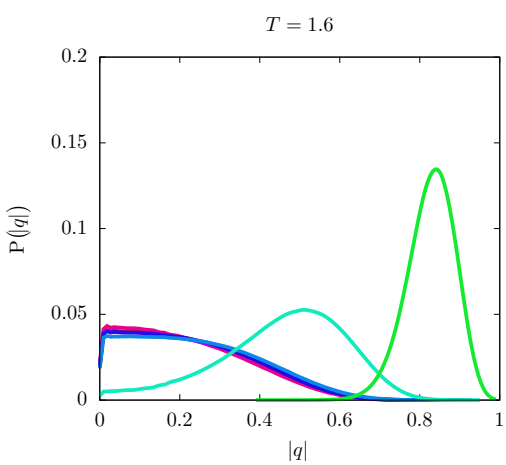
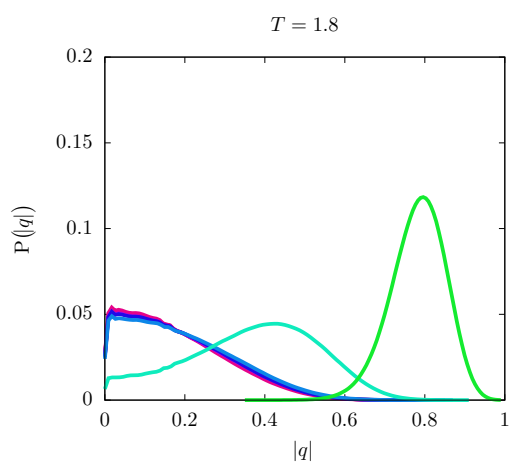


Figure 10.5



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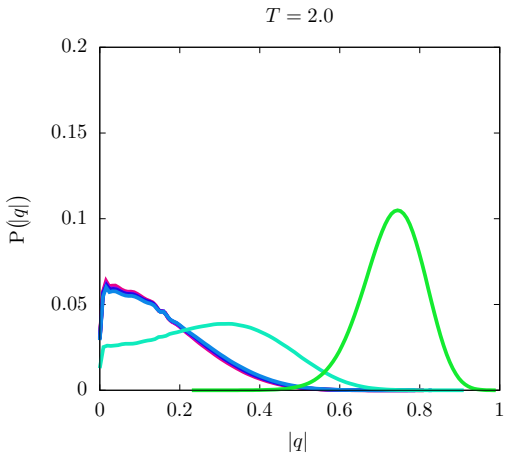

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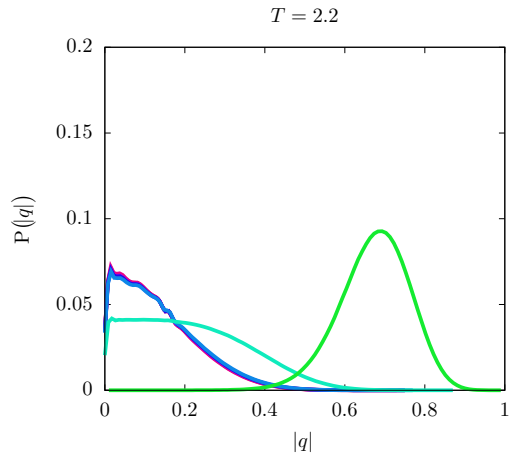

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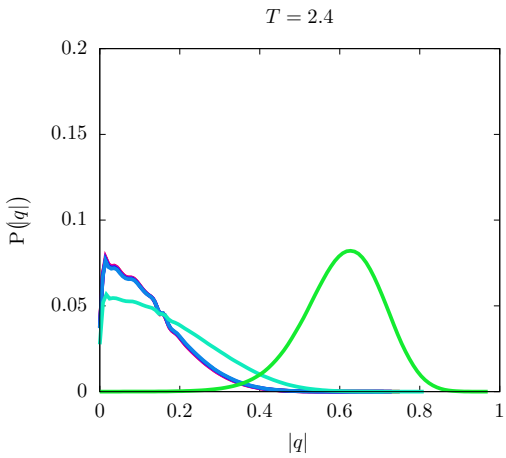

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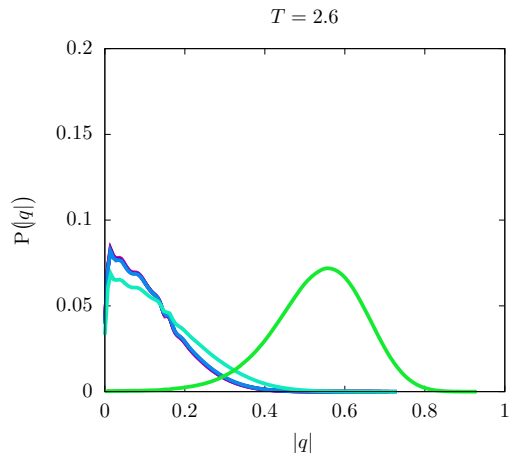

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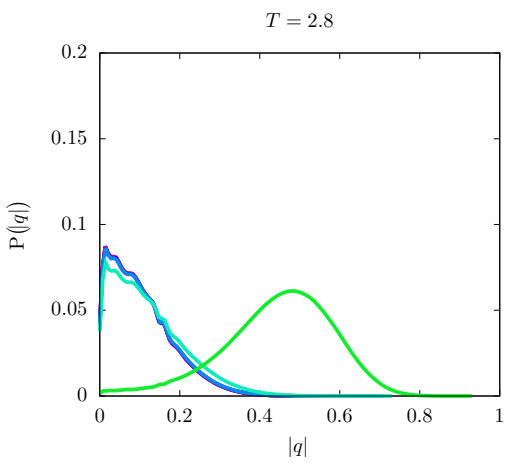


Figure 10.11

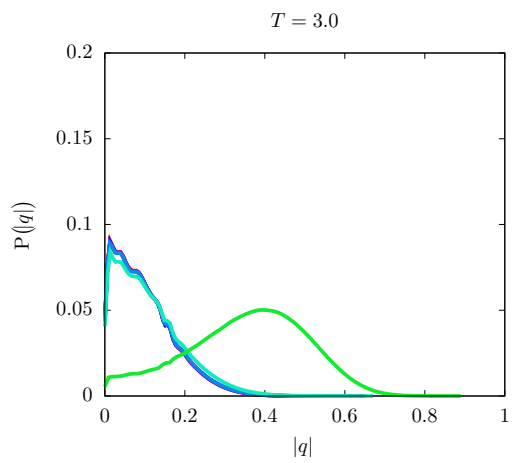


Figure 10.12

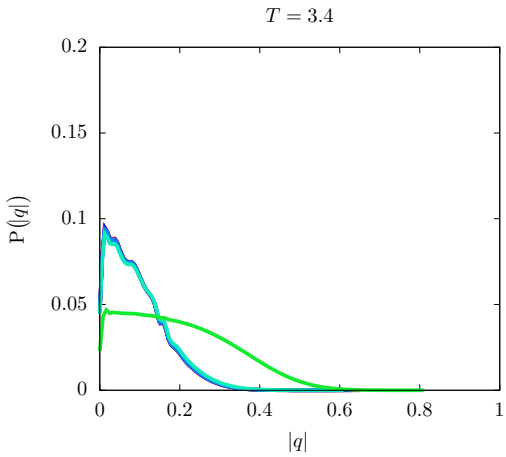


Figure 10.13

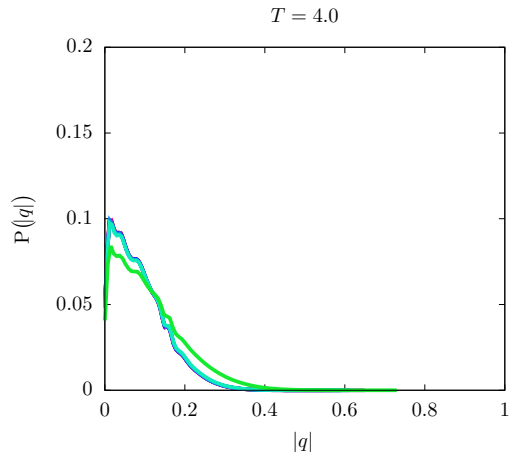


Figure 10.14

11 PL performance for different T

Considering $N = 100$, $\langle z \rangle = 4$, $N_{\mathrm{g}} = 100$, $\mathrm{C} = 400$, $\tau = 2000$ and $T_{\mathrm{F}}^{\mathrm{ini}} = -\frac{1}{3} \ln \left[\frac{1}{2} \left(1 + \tanh \frac{1}{T} \right) \right]$

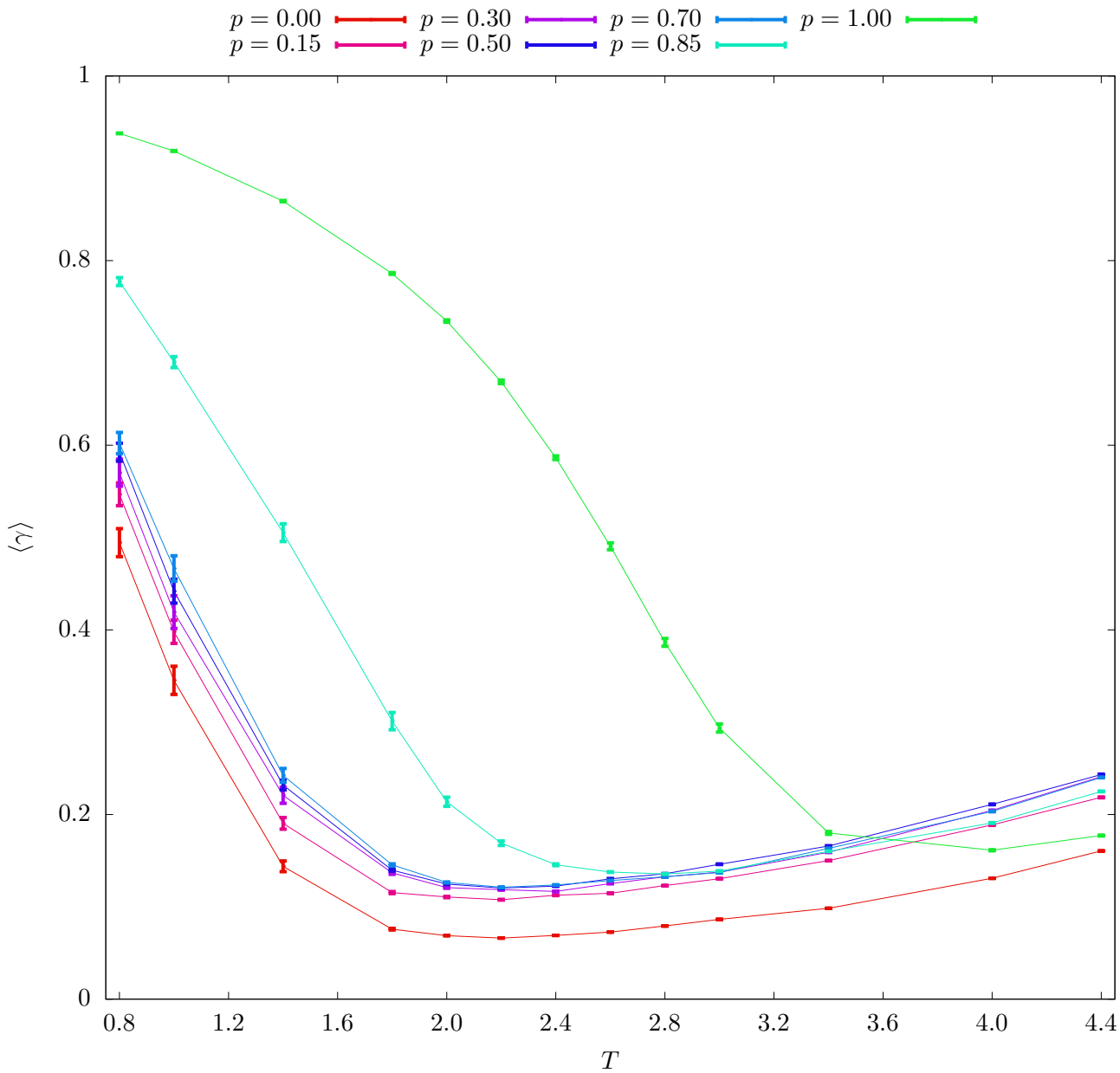


Figure 11.1: PL performance in relation to T for different p.

12 Phase Diagram

In [1] the following equations are obtained, for random ferromagnetic and anti-ferromagnetic interactions of equal strength $\pm J$ and a average connectivity $\langle z \rangle$,

$$\langle \tanh^2(J/T_{\mathrm{SG}}) \rangle_J = \frac{1}{\langle z \rangle} \qquad \langle \tanh(J/T_{\mathrm{FM}}) \rangle_J = \frac{1}{\langle z \rangle} \tag{12.1}$$

So since the pairwise distribution in this project system is

$$P(J_{ij}) = p \cdot \delta(J_{ij} - 1) + (1 - p) \cdot \delta(J_{ij} + 1) \tag{12.2}$$

and $\langle z \rangle = 4$, the relation between the different phase temperature transition and p are

$$T_{\mathrm{SG}}^{-1} = \operatorname{arctanh} \frac{1}{2} \qquad T_{\mathrm{FM}}^{-1} = \operatorname{arctanh} \frac{1}{4(2p-1)} \tag{12.3}$$

According to [2] the SG-FM transition at $T = 0$ is around $p = 0.73$, with a mixed phase up to $p = 0.8$, with this and knowing from [1] that this phases are split vertically near the multi-critical point, a full phase diagram can be drawn:

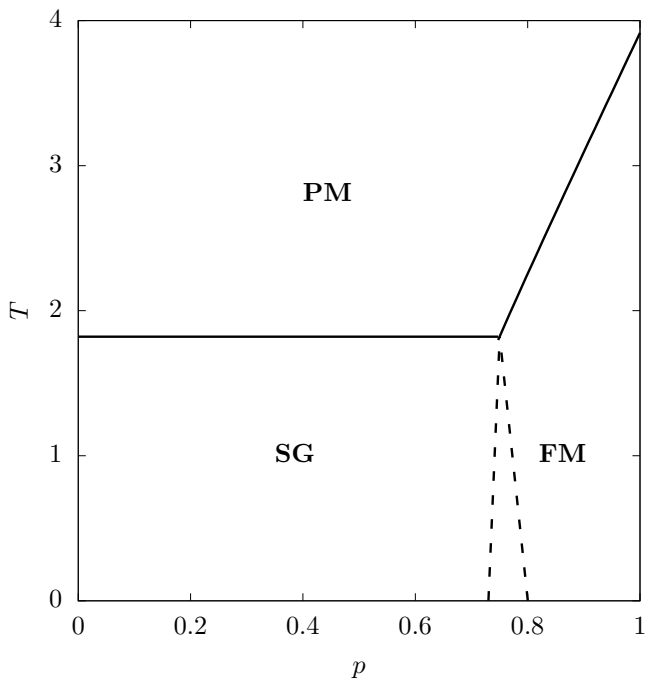


Figure 12.1

References

[1] L.Viana and A.J. Bray *Phase diagrams for dilute spin glasses*, (J. Phys. C: Solid State Phys. 18 3037 (1985)).

[2] M.O. Hase *Spin-glass behaviour on random lattices*, (J. Stat. Mech. (2012)).

13 PL performance and average magnetization per sample relation

All figures for $N = 100$, $\langle z \rangle = 4$, $T = 1.0$, $N_{\mathrm{g}} = 100$, $T_{\mathrm{F}}^{\mathrm{ini}} = 0.04$.

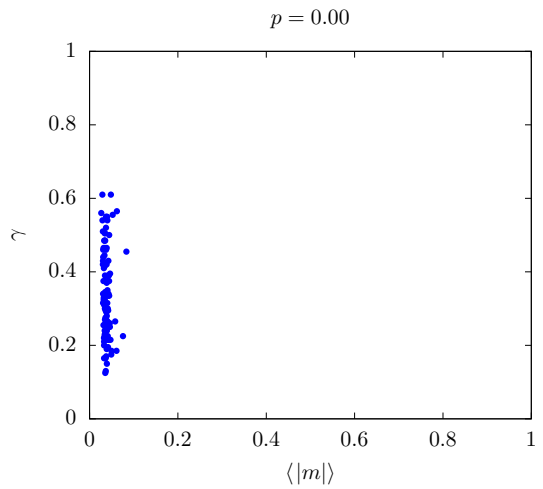


Figure 13.1

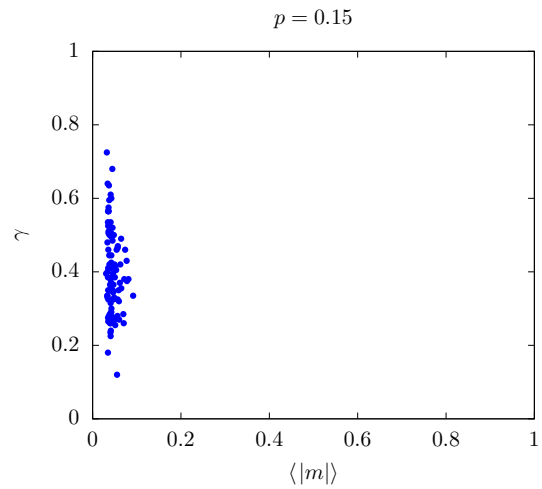


Figure 13.2

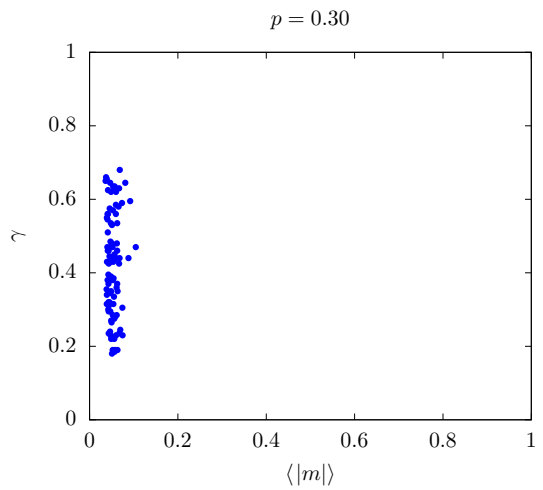


Figure 13.3

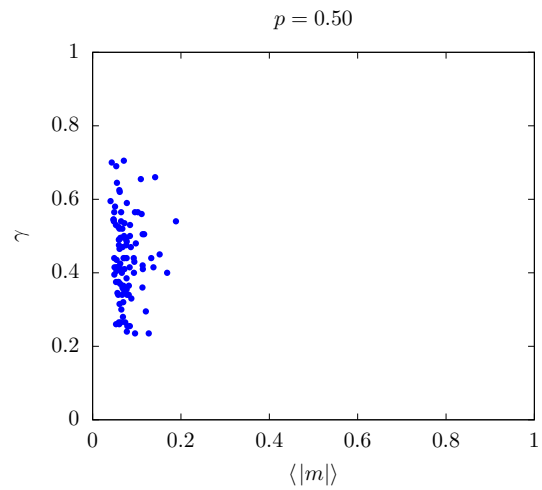


Figure 13.4

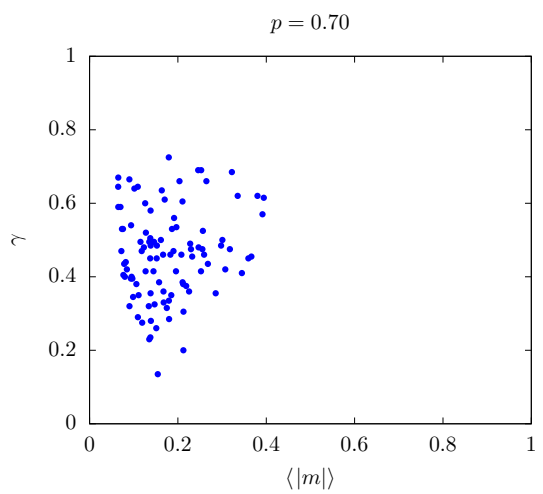


Figure 13.5

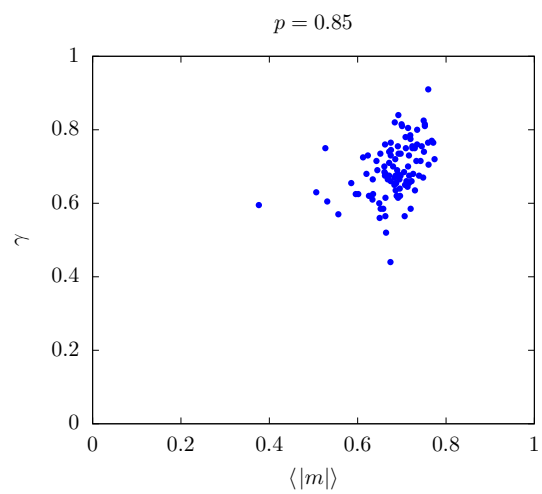


Figure 13.6

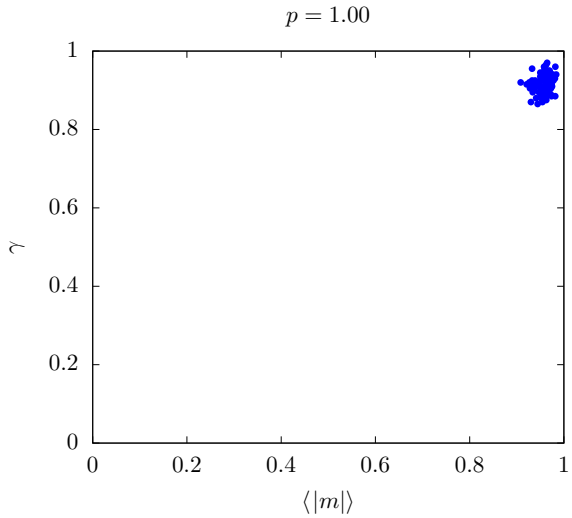


Figure 13.7

14 Average Sample Energy gamma dependency for C = 400, τ = 1000

All figures for $N = 100$, $\langle z \rangle = 4$, $T = 1.0$, $N_{\mathrm{g}} = 100$, $T_{\mathrm{F}}^{\mathrm{ini}} = 2.0$.

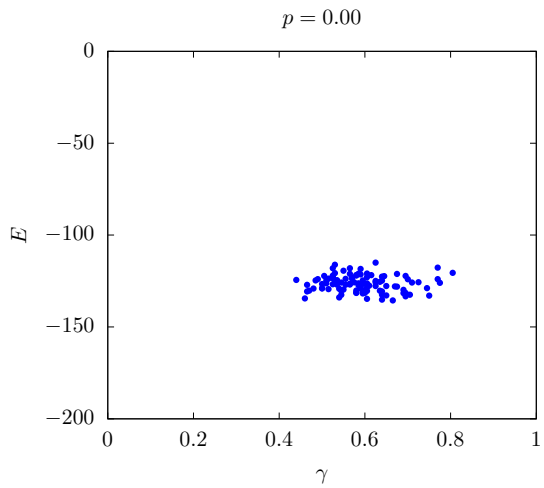


Figure 14.1

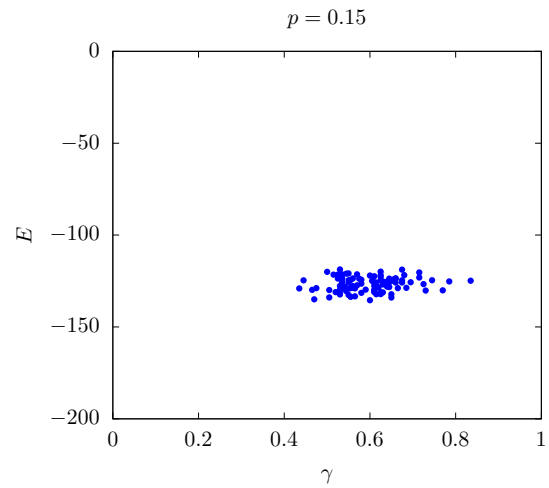
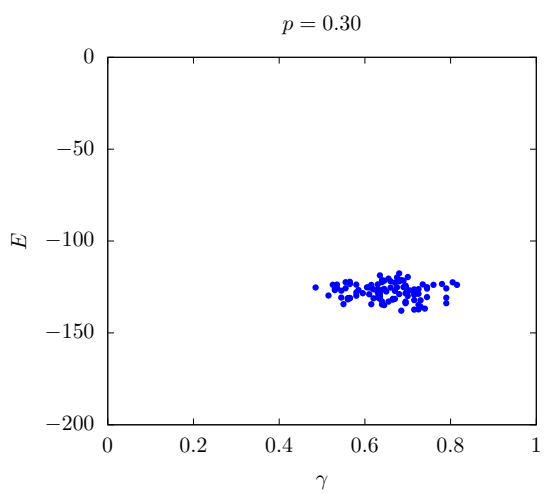


Figure 14.2



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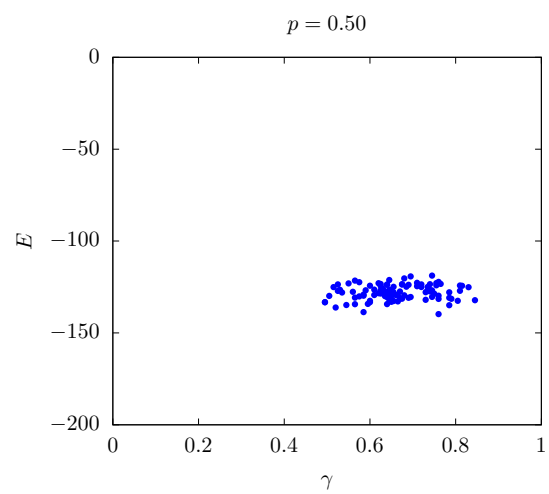

Figure 14.4

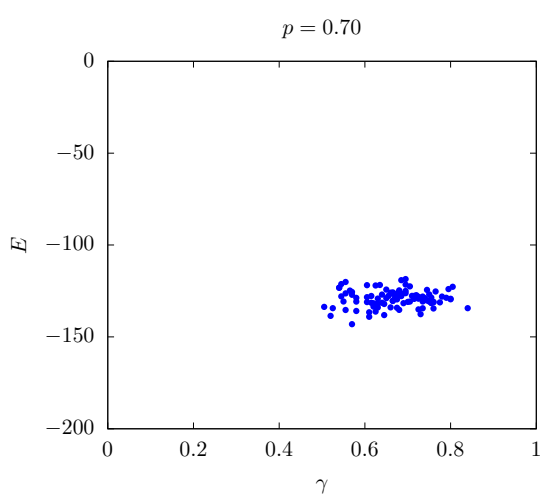

Figure 14.5

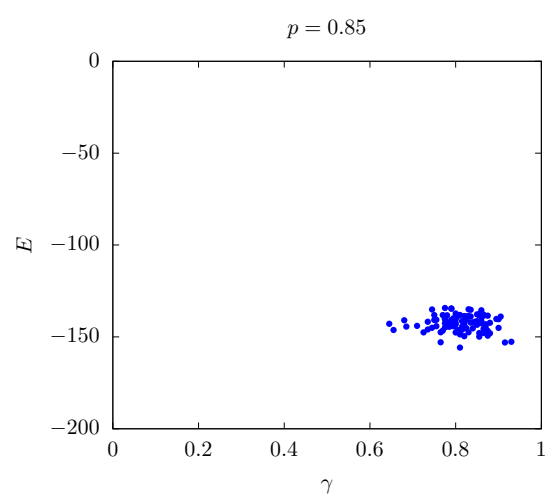

Figure 14.6

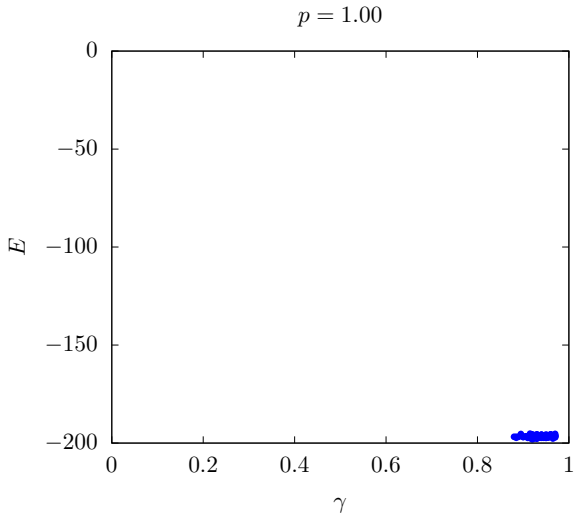

Figure 14.7